Supplementary material: When Proximity Is Not Interception: Auditing Capture Endpoints and Predictive Lane Blocking for Vehicle Pursuit in Microscopic Urban Traffic Simulation

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This supplement gives the audit details, the full tables and the secondary analyses behind the main article. Section and table numbers carry the prefix "S". Terms and abbreviations are those of the article: operational capture (OC), predictive lane blocking (PLB), the TRAINING, DEVELOPMENT and TEST cities, and the second held-out set, HELD-OUT B. The code, the analyses and the data figures reported here were produced with the AI assistance that the article describes in its section on protocol and statistics, under the same safeguards.

S-I. Persistence Audit of Operational Captures

Table S-I reports the hold audit. In the original audit, 188 of 1 433 replays had less than 60 s of follow-up because the episode horizon ended. A versioned replay extended these to 60 s. A second replay verified that each extension reproduced the original capture (same time and finisher) and that the natural and held replays agreed in position and speed at the moment the pursuers were held (188/188). With complete follow-up, no escape was observed in the 1 433 audited captures. They come from 176 scenario–evader families in seven cities and share families across methods, team sizes and seeds, so we

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report the count rather than an independence-based bound. Captures next to a red signal or a free adjacent lane are rare.

Table S-I: Hold audit of OC captures (60 s complete follow-up for all after a versioned replay of 188 censored cases).

	PLB	Ambush	Pursuit
Captures audited	562	480	391
Censored originally (< 60 s)	82	64	42
Escape, pursuers held	**0**	**0**	**0**
Red/yellow signal $\leq$40 m	2.0%	2.7%	3.1%
Free adjacent lane	1.6%	1.5%	2.0%
Released if re-planning	8.4%	14.4%	12.8%

Capture instants re-verified. The instant at which the pursuers were held came from a helper that kept a pursuer's run of valid frames open when that pursuer was absent from a frame, so that two separate fragments could be joined into one witness. A fifth audit showed this on a constructed trace. The traces of the audit had not been stored, so all 1 433 captures were replayed, and each instant was recomputed with a corrected extractor, which ends a run when the pursuer or the evader is absent or a value is invalid, and with the judge applied to each pursuer alone. Every replay reproduced the sealed instant and finisher, and the three computations agree in all 1 433 captures. No hold audit therefore started from a wrong instant, and the results above stand. The condition that breaks the earlier helper does occur in this simulator: in 4 of the 1 433 traces a pursuer is absent for 0.4 s or 2.5 s and then present again. In all four it is not the pursuer that completes the capture. No collision occurred in these episodes.

Active-escape probe. The hold audit was repeated with a probe that, from the moment of capture, requests every second a change to an adjacent lane with an onward link and a reroute through it, with safety checks on. Among the 1 433 reproduced captures, 15 escapes were observed, from 3 of 176 families. Lane-change requests were recorded in 115 episodes, which include all 15 escapes. In 1 316 episodes only reroute requests were recorded, and no escape was observed. Two episodes produced only control errors and are inconclusive for this test. The absence of a recorded lane-change request is not proof that no adjacent exit existed: the probe did not take an independent inventory of lanes at capture, and it did not verify that a

requested manoeuvre was executed. Reversing and mid-edge U-turns were not evaluated. An earlier version of this analysis presented the request groups as geometric groups; that reading is withdrawn.

S-II. Capture Events in a Published Environment

Source. We used the official repository of DGMARL-MVP [1] (`https://github.com/BUPT-ANTlab/DGMARL-MVP`, commit `62bd72e2`). Before running it we reviewed the code. The environment, which is the only part we executed, opens no network connection, evaluates no external input and deletes no file; it starts SUMO. Compiled files shipped with the repository were deleted without being run. The repository has no licence file, so we cite and evaluate it and do not redistribute it. Each worker ran a byte-identical copy of the source files (their hashes are stored with the results). Nothing in the code was edited. The only values set from outside are ones the code exposes: the path of the SUMO binary (undefined on our platform in the release), the TraCI port and the scenario name.

What the released code does. A capture is declared when the distance between the positions of a pursuer and an evader is below 5 m. In SUMO these positions are the front-bumper points. The evader is then removed from the simulation, and the episode succeeds when all evaders have been removed within 800 steps of 1 s. Evaders turn at random at each junction. Pursuers choose left, straight or right at the next junction; they have no action to stop or to wait. The network is a grid of 16 junctions with one 3.2 m lane per direction, 5 m vehicles and 200 background vehicles. A follower in the same lane is at least one vehicle length plus a gap behind its leader's front bumper, so the rule cannot be met by following; we recorded this as a prediction before the run.

Released versus reported configuration. The article does not state the capture distance; the code does. The release also differs from the reported settings: the learning rates are 10^{-12} in the release and 10^{-4} in the article, the lane speed limit is 13.9 m/s against a reported maximum of 20 m/s, and no trained weights are provided. We used SUMO 1.21; the authors' version is not stated. We therefore audit the released environment and do not reproduce the article's numbers.

Design (registered before the run). Two scenarios as released: four pursuers and two evaders (300 episode seeds) and seven pursuers and four evaders (100 seeds). Two teams, with the same seeds:

- a *null* team, in which every pursuer turns uniformly at random;
- a *chaser* team, in which every pursuer takes the turn whose next road is closest, by shortest path in the environment's own road graph, to the road of the nearest remaining evader.

Each episode ran in two modes. In the *published* mode the environment is unmodified, and an observer reads the state of the capturing pair before the evader is removed. In the *follow-up* mode the episode is identical until the first capture; the evader is then left in the simulation, and it is followed for 30 s while the pursuers keep their policy and keep it as a target. All 1 600 episodes completed. Eight episodes run beforehand as an execution check were reproduced identically. In 12 of the 577 first captures the episode horizon ended before the 30 s had passed. A versioned addendum re-ran those 12 with the horizon extended; each reproduced the same first event, so the follow-up rows of Table S-II cover all 577 first captures. The same addendum re-ran the 800 unmodified episodes to measure the stop analogue of the last row with a duration in simulation time; all 800 reproduced the original capture events.

Table S-II: Capture events in the official DGMARL-MVP environment, released configuration. P4–E2: four pursuers and two evaders; P7–E4: seven and four. Episode-level 95% bootstrap intervals.

	P4–E2		P7–E4	
	Null	Chaser	Null	Chaser
Episodes	300	300	100	100
Published success	42.7% [37.3, 48.3]	4.0% [2.0, 6.3]	39.0% [30.0, 48.0]	2.0% [0.0, 5.0]
At least one capture	90.0%	42.3%	100.0%	80.0%
Share of evaders captured	66.3%	23.2%	81.2%	34.2%
Mean steps	673	789	687	798
Capture events	398	139	325	137
same lane	0	0	0	0
opposite directions	82.9%	59.7%	82.5%	70.8%
inside a junction	17.1%	40.3%	17.5%	29.2%
headings differ by $\geq 150°$	93.7%	92.1%	91.4%	94.9%
evader faster than 1 m/s	75.6%	90.6%	70.5%	89.0%
median front-to-front distance	3.6 m	3.7 m	3.7 m	3.7 m
First captures followed for 30 s	270	127	100	80
identical to the unmodified run	270/270	127/127	100/100	80/80
+1 s: within 5 m of its captor	12.2%	23.6%	9.0%	17.5%
+5 s: within 5 m of its captor	0 of 270	0 of 127	0 of 100	0 of 80
+10 s: evader faster than 1 m/s	75.2%	91.3%	67.0%	85.0%
+10 s: over 50 m from every pursuer	94.8% [92.2, 97.4]	92.9% [88.2, 96.9]	95.0% [90.0, 99.0]	92.5% [86.2, 97.5]
+30 s: median distance to its captor	439 m	423 m	449 m	423 m
Evaders stopped $\geq$2 s behind one stopped pursuer	3.8%	2.5%	4.0%	3.2%

Reading. Table S-II shows the same pattern in both scenarios. No capture

event joined two vehicles in the same lane; more than nine in ten joined vehicles with nearly opposite headings, at a median distance close to the 3.2 m that separates the two lanes of a road. In the follow-up mode, the first capture was the same event as in the unmodified run in every episode. One second later, most evaders were already more than 5 m from the vehicle that had captured them; five seconds later, all 577 were. The share of evaders that were ever stopped for at least 2 s behind one and the same stopped pursuer, which is the closest analogue of an operational stop that this action space allows, was 2.5–4.0%, and arose from ordinary queues.

Scope. The article reports success rates of 0.77–0.91 (P4–E2) and 0.73–0.81 (P7–E4) for four learned methods, over 100 test episodes [1]. Our null team reaches about half of those values without pursuing, and a team that chases reaches almost none. Because of the configuration differences listed above, and because we did not train the published learner, this is not a reproduction and it is not evidence about that learner. It is evidence about the endpoint: in this environment the published rule is met by crossing the evader's path.

A second environment: PEPCRL-MVP. The audit was repeated on the official repository of PEPCRL-MVP [2] (commit `ac8ca695`, run unmodified with SUMO 1.21). Its release contains a real map of 53 two-way roads with one lane per direction, about 300 background vehicles, and three scenarios: 6 pursuers and 3 evaders, 7 and 4, and 8 and 5. The capture rule is that of the first environment, a distance below 5 m between vehicle positions with removal of the evader, over 800 steps. The design is the one above, with a simulation-time stop analogue and complete follow-up from the start; 150 seeds per scenario, team and mode, 1 800 episodes. One prediction was registered before the run: with 5 m vehicles, front-bumper positions and one lane per direction, no capture event would join two vehicles in the same lane.

Denominators. Each scenario and team was run with the same 150 seeds in two modes, which are separate simulations: 900 episodes with the unmodified dynamics (*published* mode) and 900 in *follow-up* mode. Of the 1 800 scheduled episodes, 1 781 completed: 884 in published mode and 897 in follow-up mode. Nineteen aborted inside the released environment, which has no branch for a pursuer that SUMO is teleporting or for an evader that has left; at most 8 of 150 in one cell, below the limit of 20% registered for interpreting a cell. Three kinds of figure therefore have three denominators. Success rates use the completed episodes of the published mode (the first row of Table S-III). The capture events are all the events of the published mode,

Table S-III: Capture events in the official PEPCRL-MVP environment (released real map, 150 episodes per team, scenario and mode). Success uses completed episodes of the unmodified mode, with episode-level 95% bootstrap intervals; capture events include those recorded before an abort; follow-up rows come from the separate follow-up mode. An episode that the unmodified environment aborted is not replaced.

	6 pursuers, 3 evaders		7 pursuers, 4 evaders		8 pursuers, 5 evaders	
	Null	Chaser	Null	Chaser	Null	Chaser
Episodes completed, of 150	150	142	149	149	150	144
Published success	31.3%	2.8%	59.1%	10.7%	58.7%	9.7%
95% interval	[24.0, 38.7]	[0.7, 5.6]	[51.0, 67.1]	[6.0, 16.1]	[50.7, 66.7]	[4.9, 14.6]
Capture events	295	138	520	314	669	444
of which before an abort	0	8	2	1	0	11
same lane	0	0	0	0	0	0
opposite directions	78.3%	82.6%	76.3%	79.0%	79.4%	75.7%
inside a junction	21.7%	16.7%	23.6%	21.0%	20.6%	24.3%
evader faster than 1 m/s	61.4%	85.5%	67.1%	79.9%	71.0%	79.5%
First captures followed for 30 s	142	96	150	143	150	146
+5 s: within 5 m of its captor	1.4%	0.0%	0.0%	0.0%	0.0%	0.0%
+10 s: over 50 m from every pursuer	79.6%	62.5%	83.3%	69.2%	82.7%	73.3%
Evaders stopped ≥2 s behind one stopped pursuer	4.4%	6.1%	6.2%	8.9%	5.3%	11.2%

2 380: 2 358 from completed episodes and 22 recorded in 13 episodes before the environment aborted them. An event before an abort is a valid event of the unmodified dynamics, and keeping it was fixed before the run; the table gives their number per cell. The follow-up figures use the first captures of the follow-up mode.

The prediction held: none of the 2 380 capture events joined two vehicles in the same lane, neither among the 2 358 of completed episodes nor among the 22 recorded before an abort (Table S-III). The null team met the published criterion in 31–59% of episodes and the chaser in 3–11%. The first capture was the same event as in the unmodified run in 814 of 814 checked episodes; all 827 first captures have 30 s of follow-up, and 2 of the 827 evaders were within 5 m of their captor after 5 s. After 10 s, 76% were more than 50 m from every pursuer, fewer than on the grid: the map is smaller and the teams larger. The article reports success rates of 0.68, 0.62 and 0.56 for its method in the three scenarios, and 0.27–0.62 for eight baselines, over 100 tests; in the two larger scenarios the null team is at the level of the proposed method. As before, the released configuration may differ from the reported one, a later SUMO version was used and the learner was not trained, so this is evidence about the endpoint and not about the learner. The two audited environments come from one research group and share code. The second adds a real map and a journal article; it does not add an independent group.

S-III. Other Controllers and Full Tables

Cut. A further heuristic controller assigns distinct gates to the pursuers by urgency. On real cities, pursuit and cut perform alike. Against the optimal evader, pursuit is better by 0.08–0.10. For pursuit and cut teams, the excess over the independence reference is at most zero in all ten cells. On literature-like grids, cut beats pursuit by +0.29 [0.19, 0.38].

Junction ambush. The junction ambush beats pursuit by +0.04 to +0.12, but a full ambush was feasible in only 8–18% of decision steps, which motivated pre-positioning.

Held-out evaluation. Table S-IV is the full table of the single-shot evaluation, including the two controllers of the first MAPPO study (A), which imitate PLB. They show no detected difference from it: KAN −0.001 [−0.040, 0.037] and MLP −0.009 [−0.048, 0.030]. No cell exceeds the independence reference.

Table S-IV: Single-shot evaluation on the four TEST cities with frozen methods. OC capture rates. 7 200 team episodes were scheduled for the five methods. 210 failed scenario placement, in 42 conditions failing identically for all five methods, leaving 6 990 completed. Single-pursuer and certificate-replay runs are reported separately.

Evader	Team	PLB	Junction ambush	Pursuit	MAPPO-KAN (A)	MAPPO-MLP (A)
Random	3	0.299	0.175	0.162	0.299	0.305
	5	0.351	0.260	0.221	0.396	0.377
	7	0.370	0.292	0.292	0.448	0.468
Avoidant	3	0.297	0.203	0.127	0.259	0.241
	5	0.373	0.253	0.171	0.323	0.316
	7	0.367	0.266	0.215	0.329	0.304
Optimal	3	0.253	0.156	0.130	0.247	0.234
	5	0.260	0.188	0.091	0.273	0.286
	7	0.364	0.260	0.143	0.338	0.299

Second held-out set (HELD-OUT B). After every method and analysis had been frozen, four further cities were named, with four reserves and the acceptance criteria, before any map was downloaded: Denver, Vancouver, Barcelona and Buenos Aires, each a square of 2.4 km centred on a stated point. The networks were built from OpenStreetMap with the road types and conversion options recorded in the headers of the existing networks. A city was accepted if it met the usability thresholds of the existing maps and the frozen generator placed a team of seven in at least 15 of 20 scenarios; all

four were accepted. One deviation occurred before any controller ran and was corrected: a server timeout made the build script substitute the first reserve for Barcelona, which is not a registered reason for rejection; the download was repeated and the registered city was used. The evaluation was a single shot with PLB, the junction ambush and pursuit, the avoidant, random and optimal evaders, teams of 3, 5 and 7, 20 scenarios and two SUMO seeds, plus the null team of five pursuers that drive random routes. Of 4 640 scheduled episodes, 4 498 were valid; the 142 failures are placements that fail identically for the three methods. No collision occurred. Table S-V gives the results.

Table S-V: HELD-OUT B (Denver, Vancouver, Barcelona, Buenos Aires), single shot. Rates pooled over the avoidant and random evaders and the three team sizes; null team of five pursuers. Contrasts with family-clustered intervals (157 families).

	OC	Body gap ≤5 m	Centre ≤5 m
PLB	0.291	0.792	0.647
Junction ambush	0.267	0.777	0.641
Pursuit	0.227	0.776	0.643
Null team	0.016	0.273	0.214

Registered primaries (97.5%)
PLB minus pursuit, OC: +0.064 [0.015, 0.114]; over cities and families (95%) [0.009, 0.132]
Body gap ≤5 m minus OC, pursuit team: +0.549 [0.476, 0.620]
Secondary contrasts (95%)
PLB minus pursuit, optimal evader: +0.073 [0.017, 0.131]
PLB minus junction ambush: +0.024 [−0.017, 0.064]
PLB minus pursuit under body gap ≤5 m: +0.016 [−0.002, 0.037]; under centre ≤5 m: +0.004 [−0.016, 0.025]
PLB minus pursuit by team size: +0.054, +0.082, +0.057
By city: Denver +0.145 [0.033, 0.254]; Vancouver +0.029 [−0.075, 0.121];
Barcelona +0.031 [−0.014, 0.086]; Buenos Aires +0.050 [0.000, 0.111]

Both registered primaries were met. The measurement findings replicate on cities that did not exist in the project when everything was frozen: the proximity rule exceeds OC by 55 points for the pursuit team, the null team scores 1.6% on OC against 21–27% on the proximity rules, and the advantage of PLB under OC is not visible under either proximity rule. The controller findings replicate in part. PLB beats pursuit, by 6.4 points against 14.5 on TEST; a post hoc comparison of the two sets, each resampled over its own families, gives a difference of +0.081 [0.011, 0.149]. PLB is not distinguishable from the junction ambush on this set, whereas on TEST it was (+0.102); the difference between the sets is +0.078 [0.015, 0.140]. The gap between the body-gap rule and OC is also smaller (+0.549 against +0.666 with the same pooling; difference +0.117 [0.028, 0.206]). Capture differs strongly between networks: the OC rate of the pursuit team is 0.02 in Buenos Aires, 0.10

in Barcelona, 0.34 in Denver and 0.43 in Vancouver. We did not identify what makes a network easy or hard, and we report the two held-out sets side by side without pooling them, as registered. This evaluation was judged with the invalid-value check of Section S-VII in place; a re-run of 140 earlier episodes with that check reproduced their recorded outcomes.

Fresh scenarios on the seven development-phase cities. Tables S-VI and S-VII give the per-cell rates behind the pooled contrasts of the article. Against the goal-directed fugitive, single PLB pursuers are weak (0.05), and the team exceeds the independence reference at every size (+0.05 to +0.08), although not significantly after Bonferroni adjustment.

Table S-VI: PLB on fresh scenarios, seven development-phase cities. OC capture rates.

Evader	Team	PLB	Ambush	Pursuit
Avoidant	3	0.279	0.217	0.170
	5	0.361	0.307	0.223
	7	0.369	0.336	0.223
Random	3	0.307	0.200	0.200
	5	0.344	0.311	0.263
	7	0.404	0.393	0.359

Table S-VII: PLB versus pursuit against the optimal evader (7 development-phase cities) and a goal-directed fugitive that PLB's predictor does not model (7 cities).

Setting	Team	PLB	Ambush	Pursuit
Optimal	3	0.425	0.250	0.187
	5	0.357	0.353	0.180
	7	0.440	0.376	0.214
Goal-directed	3	0.196	0.115	0.074
	5	0.237	0.196	0.148
	7	0.248	0.230	0.144

S-IV. Learning Studies

Option PPO. Option PPO with MLP or KAN heads, behaviour-cloned from the junction-ambush rule and fine-tuned with circuit breakers, matched

the rule it was cloned from (difference 0.000 [−0.004, 0.004]). Our KAN layer reproduces the reference implementation (pykan 0.2.8) to 10^{-5}.

The controller. A centralized-critic PPO in the style of MAPPO uses a shared actor (MLP or KAN) and a critic that sees a seven-feature global state; each pursuer chooses a blocking lane, chase or stay every 3 s. The actor is sequential: pursuers decide in a fixed order within a step, and the targets chosen earlier in the step are visible to those that follow. Earlier documentation described the agents as deciding simultaneously; that description was wrong and has been corrected.

An implementation defect found by audit. In the first three learning studies (first MAPPO, MAPPO-v2 and the proximity-training study), the global state given to the critic was built after the agents of the step had chosen: two of its features, the coverage of the next two junctions, read the targets just chosen. The value subtracted as a baseline therefore depended on the actions it was compared with, which biases the policy gradient. In addition, actions were sampled with a global random generator, so a training rollout could not be reproduced from its declared seed. Neither defect touches behaviour cloning, which uses neither the critic nor sampling, nor the evaluation of a stored policy, which is greedy. The rates reported for stored policies are therefore valid descriptions of those policies. What the defects remove is any conclusion about how a correct implementation behaves under fine-tuning. Unit tests now reproduce both defects in the old code and check their absence in the corrected controller (MAPPO-v3).

First MAPPO study. In the first study, fine-tuning lowered validation capture, and the imitation checkpoints were kept. Besides the implementation defect, that study used one seed and had two design defects: its sampler covered only 4 of the 16 city–setting combinations, and it optimized a looser "trapped" proxy rather than OC.

Second study (MAPPO-v2). The second study addresses the two design defects, but not the implementation defect, which was found later:

- it covers all 16 combinations;
- it optimizes and selects on OC, with episodes truncated at capture;
- it uses a teacher faithful to the planner;
- it trains two seeds per head;
- its training update has unit tests.

Fine-tuning again lowered validation capture in all four runs, from 0.36–0.37 to 0.20–0.22 within 2 048 episodes, and the breakers kept the imitation checkpoints. On fresh scenarios (15 120 scheduled episodes), the pre-registered primary was not met: MLP minus PLB is +0.025 and KAN minus PLB +0.017, with 97.5% family intervals [−0.011, 0.059] and [−0.015, 0.048]. The selected checkpoints are imitation checkpoints, kept because fine-tuning reduced validation capture. Per seed, the 95% family intervals for v2 minus PLB range from [−0.010, 0.044] to [0.000, 0.058]; these are conditional on the selected checkpoints and do not measure variability between training runs. The faithful teacher itself scores +0.024 [−0.002, 0.051] over the 1 s planner, consistent with its 3 s decision commitment. Because of the implementation defect, the decline under fine-tuning in these four runs cannot be attributed to the algorithm.

Repetition with a corrected controller (MAPPO-v3). The corrected controller records the critic's state before any agent of the step has chosen, and samples with a generator seeded per episode. The study was registered before it ran. It starts from the four stored imitation checkpoints of MAPPO-v2 and has three arms, each run for two seeds:

- *F*, fixes only: fine-tuning as in MAPPO-v2 with the corrected controller, for the MLP and KAN heads;
- *W*, fixes and critic warm-up: as F, preceded by 256 sampled episodes on which only the critic is fitted, because the imitation checkpoints carry an untrained critic (MLP head);
- *P*, proximity reward: reward at the first body gap of at most 5 m, with a fixed budget of 2 048 episodes (MLP head).

Hyperparameters, validation and breakers are those of MAPPO-v2. Table S-VIII gives the validation curves and the test rates. In the six runs with OC as the reward, validation capture stayed within 0.02 of its start over 2 048 episodes; in MAPPO-v2 the same schedule had taken it from 0.36–0.37 to 0.20–0.22. No run reached the registered gain of 0.02, so every selected checkpoint is its imitation start.

The test used fresh scenarios on the seven development-phase cities (avoidant and random evaders, teams of 3, 5 and 7, 20 scenarios and two SUMO seeds). Of 23 520 scheduled episodes for 14 policies, 616 failed scenario placement, in 44 conditions that fail identically for all policies, leaving 22 904 (273 families). Neither registered primary was met:

Table S-VIII: MAPPO-v3. Validation success at the start and after 2 048 episodes (OC for arms W and F, proximity for arm P; two seeds), and test rates of the imitation starts and of the final policies on fresh scenarios (1 636 episodes per policy, averaged over the two seeds).

	Validation, start → end		Test, final policies	
Arm	Seed 1	Seed 2	OC	Body gap ≤5 m
Imitation start, MLP	–	–	0.418	0.918
Imitation start, KAN	–	–	0.421	0.920
W: fixes and critic warm-up, MLP	0.474 → 0.468	0.474 → 0.463	0.433	0.919
F: fixes only, MLP	0.474 → 0.474	0.474 → 0.468	0.427	0.918
F: fixes only, KAN	0.468 → 0.484	0.463 → 0.463	0.424	0.919
P: proximity reward, MLP	0.937 → 0.932	0.937 → 0.932	0.426	0.918
PLB	–	–	0.400	0.920
Pursuit	–	–	0.282	0.925

- *Primary 1*, a validated improvement on imitation (arm W, selected checkpoint minus its start): the difference is zero, because the breaker kept the start in both seeds.
- *Primary 2*, a widening of the gap between endpoints under proximity training (arm P, final policies against their starts): −0.009 [−0.022, 0.005].

The final policies differ from their starts in OC by +0.015 [0.003, 0.028] in arm W, +0.009 [−0.005, 0.025] in arm F with the MLP head, +0.003 [−0.011, 0.017] in arm F with the KAN head and +0.009 [−0.005, 0.022] in arm P. The interval of arm W is above zero; this is a secondary contrast of 1.5 points, below the gain the registered rule required in validation, and we do not claim it as an improvement. The warm-up made no detected difference (W minus F, final policies: +0.006 [−0.007, 0.019]).

We read these results as follows. The decline reported for the first implementation did not recur after the critic input and the sampling were corrected and the study was repeated. The design does not isolate why. The two corrections were made together; the training, validation and test scenarios of MAPPO-v3 are new; and arm W adds a critic warm-up. Even the validation capture of the same imitation starts differs between the two studies (0.36–0.37 against 0.46–0.47), because the validation scenarios differ. We therefore do not attribute the earlier decline to a specific cause; an earlier version of this supplement did, and that statement is withdrawn. In MAPPO-v3, no

consistent deterioration was observed and no validated gain was obtained at this budget. Four limits apply. Two seeds per arm describe these runs and do not estimate the variability between training runs. The imitation policies are nearly deterministic (the mean entropy of the sampled policy is about 0.02 nat), so fine-tuning explored little at the registered entropy coefficient, and the learner was not tuned further. A difference that is not detected is not a demonstration of equivalence. And the training loop of MAPPO-v3 still carried the two defects described next.

Erratum of the registered protocol. The protocol of MAPPO-v3 states 840 test episodes per policy. The registered design (7 cities, 20 scenarios, 2 SUMO seeds, 2 evaders and 3 team sizes) and the execution are 1 680 scheduled episodes per policy, of which 1 636 completed. The figure in the protocol is an arithmetic error; the sealed protocol is not edited.

Two defects in the training loop, found by a later audit. A fifth audit examined the training loop that every learning study shares, MAPPO-v3 included, and found two defects in it.

First, a decision was trained whenever its time did not exceed the end of the trained part of the episode. The decision taken at the 300 s horizon has no future inside the episode; it was nevertheless trained, with a time cost, in every episode that did not reach the training objective. And when a capture completed exactly on a 3 s decision tick, the decision taken at that instant, after the capture was complete, received the reward of the capture. Second, the collision penalty used an episode-level flag. A collision at any time of the simulated episode, including after the capture at which training was cut, changed the reward of the last trained decision from $+0.995$ to -0.005.

The retained records bound the incidence only in part, because the weights of the training iterations were not stored and training episodes cannot be replayed:

- The log of MAPPO-v3 stores three rounded rates per iteration. They determine the number of valid episodes uniquely in 192 of 256 iterations; overall, between 13 931 and 16 026 of the 16 384 attempted training episodes were valid, and between 6 492 and 7 399 did not reach the training objective. Each of these that ran to the horizon trained one decision without a future, about one in a hundred of its decisions.

- In the MAPPO-v2 test, whose records store the capture instant, 261 of 4 752 captures by the learned controllers fall on a decision tick (5.5%).

- No collision was recorded in the 22 904 test episodes of MAPPO-v3, in the 11 466 of the proximity study or in the 9 984 of the first study, and 8 were recorded in the 14 688 of MAPPO-v2. Collisions during training were not recorded per episode.

- The first training batch of every MAPPO-v3 run was sampled with a stored imitation start, so it can be replayed. Replaying these 512 episodes reproduces exactly the three rates logged for the first iteration of all eight runs. In the 499 valid episodes the earlier loop trained 31 433 decisions, of which 201 (0.6%) had no future inside the episode. They are spread over 201 episodes (40%): 187 at the horizon, 11 at a capture that fell on a decision tick (of 148 captures) and 3 at a first proximity that fell on a tick. No collision occurred, and no capture instant differs between the earlier helper and the corrected extractor.

These defects do not touch behaviour cloning, which uses pairs of observation and teacher choice and no reward, so the imitation starts are unaffected. They do not touch the evaluation of a stored policy either: it is greedy, and the judge scores the whole trace. Every rate in Table S-VIII is therefore a valid measurement of a stored policy. What the defects affect is how the fine-tuned policies were produced: the value targets and the advantages of the last decisions of an episode. The comparisons between final policies and their starts in the studies up to MAPPO-v3 therefore describe policies produced by a training loop with known defects. The repetition reported below (MAPPO-v4) was run to see whether their conclusions survive the correction.

Corrected training loop. The corrected loop builds explicit transitions: an observation, the action taken, the interval during which it was in force, the reward earned in that interval, and whether the trained part of the episode ends there. A decision is trained only if it is taken strictly before the end; a decision at or after the end is kept in the record as a terminal observation and never trained. The reward of the objective goes to the transition in whose interval the event occurs. The time cost is proportional to the duration of the interval. The collision penalty is applied once, to the transition in which the first collision inside the trained part occurred; later collisions are recorded with their time and not penalized. The capture instant comes from the corrected extractor of Section S-I. The loop writes a ledger with one row per attempted episode (identifiers and seeds, end reason, capture and proximity instants, timed collisions, decisions recorded and trained, the reason for any exclusion and whether the episode entered the update), counts per update,

statistics averaged over every minibatch, and the weights of every iteration, so that any training batch can be replayed. Unit tests reproduce each defect in the earlier code and test the corrected loop in the cases that the audit listed: a capture exactly on a tick, between ticks and at the horizon, an objective already met at the first decision, and collisions before, on and after the cut. Given the rewards of the earlier contract, the corrected update reproduces the earlier update exactly, so the two loops differ in which decisions are trained and in their rewards, and in nothing else.

Repetition with the corrected training loop (MAPPO-v4). The study was registered before it ran, with a design fixed before the decision to run it: the controller, arms, heads, seeds, order, hyperparameters, breakers, budgets and imitation starts of MAPPO-v3, and its training and validation scenarios, so that the first batch of every run is the batch that MAPPO-v3 sampled. The test scenarios are new. The two primaries are those of MAPPO-v3. Three integrity checks were fixed in the protocol and all held: the validation value of every imitation start equals the one that MAPPO-v3 logged (8 of 8 runs), the first batch of every run has the counts of the reconstruction described above (8 of 8), and the 73 recorded source files were unchanged at the end.

Training ledger. Budgets are counted in attempts. Each run made 2 048 training attempts, of which 1 997 to 2 009 were valid episodes. Over the eight runs, fine-tuning used 16 384 attempted episodes, of which 358 failed scenario placement and 16 026 were valid; all valid episodes entered an update (970 931 transitions). The critic warm-up, which exists in arm W only and therefore for the MLP head only, used 512 attempted and 503 valid episodes. Validation used 7 680 scheduled episodes, 192 per validation, of which 80 failed placement, which leaves 190 valid episodes in every validation. Of the 16 529 valid sampled episodes, 8 869 ended by the training objective, 5 833 at the horizon and 1 827 because the evader left the network. No collision occurred in any training, validation or test episode, and no episode was excluded as unresolved. With these episodes the earlier loop would have trained 1 012 162 decisions, of which 6 281 (0.6%) had no future inside the episode, in 38% of the episodes; in the runs rewarded with OC, 315 of 5 140 captures (6.1%) fell on a decision tick. In 9 sampled episodes a vehicle was absent inside its record, in one of them the pursuer that completes the capture; the earlier helper and the corrected extractor give the same instant in all 6 762 captures of training and in all 8 521 of the test.

Results. In the six runs rewarded with OC, validation capture stayed between 86 and 92 of 190 valid validation episodes; no run reached the registered

Table S-IX: MAPPO-v4 (corrected training loop). Validation successes among 190 valid validation episodes (192 scheduled) at the start and after the 2 048 training attempts of a run (OC for arms W and F, proximity for arm P; two seeds), and test rates of the imitation starts and of the final policies on new scenarios (1 594 episodes per policy, averaged over the two seeds).

	Validation, start → end		Test, final policies	
Arm	Seed 1	Seed 2	OC	Body gap ≤5 m
Imitation start, MLP	–	–	0.392	0.896
Imitation start, KAN	–	–	0.387	0.896
W: critic warm-up, MLP	90 → 89	90 → 88	0.403	0.895
F: no warm-up, MLP	90 → 90	90 → 88	0.393	0.894
F: no warm-up, KAN	89 → 92	88 → 90	0.397	0.896
P: proximity reward, MLP	178 → 177	178 → 178	0.389	0.895
PLB	–	–	0.376	0.893
Pursuit	–	–	0.248	0.897

gain of 0.02, so every selected checkpoint is its imitation start (Table S-IX). Of 23 520 scheduled test episodes for 14 policies, 1 204 failed scenario placement, in 86 conditions that fail identically for all policies, leaving 22 316 (267 families). Neither registered primary was met:

- *Primary 1*, a validated improvement on imitation (arm W, selected checkpoint minus its start): the difference is zero, because the breaker kept the start in both seeds.
- *Primary 2*, a widening of the gap between endpoints under proximity training (arm P, final policies against their starts): +0.003 [−0.008, 0.014].

The final policies differ from their starts in OC by +0.011 [−0.005, 0.026] in arm W, 0.000 [−0.012, 0.013] in arm F with the MLP head, +0.010 [−0.001, 0.021] in arm F with the KAN head and −0.004 [−0.014, 0.007] in arm P. No interval is above zero; the interval of arm W, which was above zero in MAPPO-v3, is not above zero here. The warm-up made no detected difference (W minus F, final policies: +0.010 [−0.002, 0.022]). On this third set of fresh scenarios, PLB minus pursuit is +0.128 [0.084, 0.170] under OC, −0.004 [−0.011, 0.003] under the body-gap rule and −0.021 [−0.038, −0.006] under the centre-distance rule.

Reading. With the corrected loop, again no consistent deterioration was observed and no validated gain was obtained, and under proximity training no change of OC or of the gap between endpoints was detected. The intervals include small losses as well as small gains; no test of equivalence or non-inferiority was made. The two registered outcomes of MAPPO-v3 are therefore unchanged by the correction of the loop. This is not a demonstration that the defects had no effect: the two studies use different test scenarios, two seeds per arm do not estimate the variability between training runs, and a difference that is not detected is not a demonstration of equivalence. The sampled policies remain nearly deterministic (mean entropy 0.02–0.03 nat), so fine-tuning explored little, and the learner was not tuned. The corrected loop has unit tests for the cases that the audit listed and a complete ledger, which does not prove that no other defect exists. MAPPO-v3 is not replaced: both studies are reported, and MAPPO-v3 remains labelled as trained with a loop that had the two defects.

The imitation controllers against PLB. Table S-X lists every test set on which an imitation MAPPO controller (MLP head) was compared with PLB. The difference ranges from −0.009 to +0.047, and its interval is above zero in two of the six sets, one of which was analysed post hoc. The statement that no advantage was detected, made in an earlier version, was therefore too strong. The cause of the difference is not established. One candidate, holding each decision for 3 s, was tested in a registered replication and not confirmed (+0.010 [−0.012, 0.033]).

Table S-X: Imitation MAPPO (MLP head) minus PLB, OC capture, on every test set where both were run.

Test set	Difference	Status
First MAPPO study, own fresh scenarios	+0.047, interval above zero	registered primary, met
Held-out TEST cities	−0.009 [−0.048, 0.030]	registered secondary
MAPPO-v2, own fresh scenarios	+0.025 [−0.011, 0.059] (97.5%)	registered primary, not met
Proximity-training study, fresh scenarios	+0.043 [0.016, 0.071]	post hoc
MAPPO-v3, fresh scenarios	+0.017 [−0.014, 0.049]	registered secondary
MAPPO-v4, fresh scenarios	+0.017 [−0.012, 0.046]	registered secondary

Training on the proximity endpoint (first implementation). To mimic the usual setting of this literature, the MAPPO-v2 controller (MLP head) was fine-tuned from the same imitation start with the proximity endpoint as its reward: +1 at the first instant a pursuer is within 5 m (body gap) of the evader, with the episode truncated there. Each of two seeds ran a fixed

2 048 episodes without early stopping, as registered before the run, because proximity success is already above 0.9 after imitation and a validation breaker would have kept the imitation checkpoint. Validation proximity stayed at 0.91–0.92 throughout.

The test used fresh scenarios on the seven development-phase cities (avoidant and random evaders, teams of 3, 5 and 7, 20 scenarios and two SUMO seeds). Of 11 760 scheduled episodes for seven controllers, 294 failed scenario placement, in 42 conditions that fail identically for all controllers, leaving 11 466 (273 families). Every episode was scored under OC and under the proximity rules (Table S-XI).

Table S-XI: Proximity-training study. Rates over 1 638 episodes per controller.

Controller	OC	Body gap ≤ 5 m	Centre ≤ 5 m
MAPPO, selected on OC	0.440	0.925	0.860
PLB	0.397	0.921	0.840
MAPPO, trained on proximity (final policies)	0.278	0.933	0.858
Pursuit	0.274	0.934	0.861

The registered primary is the gap between endpoints (body-gap proximity minus OC) of the final proximity-trained policies, minus the same gap for the controller selected on OC. It is +0.170 (family 95% interval [0.135, 0.205]), so the primary was met. In OC, the proximity-trained policies are −0.162 [−0.197, −0.127] below the controller selected on OC (−0.164 and −0.160 per seed) and −0.118 [−0.153, −0.085] below PLB. In proximity success they differ from the controller selected on OC by +0.008 [−0.001, 0.020]. PLB minus pursuit is +0.123 [0.088, 0.160] under OC, −0.013 [−0.025, −0.004] under the body-gap rule and −0.021 [−0.040, −0.002] under the centre-distance rule.

Four points limit the reading. First, these policies were trained with the defective implementation, and in the repetition with the corrected controller (arm P above) the loss of OC did not recur: proximity training left OC unchanged and the gap between endpoints did not widen. The repetition does not isolate what produced the loss. This study is therefore not evidence about proximity rewards, and the article makes no such claim. What remains true concerns these two stored policies only: one is 16 points worse on OC, and a monitor based on proximity does not show it. Second, the comparison is between a selected checkpoint and final policies, as registered. The best-validation checkpoints of the proximity runs are their imitation

checkpoints, and with them the difference in gaps is +0.007 [−0.010, 0.024]. Third, a difference that is not detected is not a demonstration of equivalence. Fourth, the proximity-trained controller is our own implementation and not a published method; Section S-II audits a published environment.

S-V. Exploratory: KAN, Spline, MLP and Logistic Models of Capture

This analysis is post hoc. We fit four models of capture on six start-state features, using TRAINING cities, and evaluate them on held-out DEVELOPMENT and TEST episodes: a KAN (pykan), a spline-additive logistic model, an MLP selected by grouped cross-validation, and a logistic regression. Fig. S-1 compares, on the same held-out rows and bins, the mean predicted probability with the observed capture rate. DEVELOPMENT and TEST are evaluated separately. Uncertainty comes from a bootstrap over true scenario families: all team-size and speed variants of a scenario are resampled together, giving 230 families in DEVELOPMENT and 156 in TEST. We report the observed estimate and the interval separately.

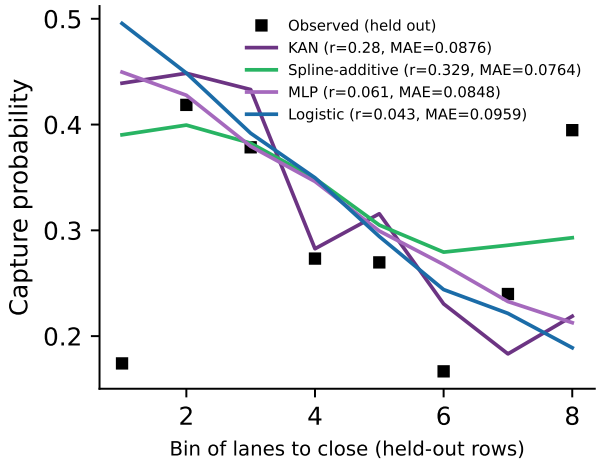


Figure S-1: Observed held-out capture by bin of lanes to close, and the mean prediction of each model for the same rows (matched estimand; DEVELOPMENT and TEST rows pooled; descriptive). Post hoc.

On TEST, the AUC is 0.597 for the KAN, against 0.645 (spline-additive), 0.647 (MLP) and 0.646 (logistic). The observed KAN differences are about

−0.05, with family 95% intervals of roughly [−0.11, 0.01]. On DEVELOPMENT, AUCs are 0.573, 0.575, 0.553 and 0.566, and every interval for a KAN difference includes zero. Bin-level correlations with observed capture across lanes to close are, on TEST, 0.68, 0.80, 0.65 and 0.63, and on DEVELOPMENT, 0.28, 0.29, 0.10 and 0.02. The observed KAN minus spline-additive differences are −0.12 and −0.01, and all their intervals include zero. No KAN advantage was detected, in line with controlled comparisons of KANs and MLPs [3]; this does not establish equality. Pruning and symbolic distillation of the KAN give a near-linear logit dominated by lanes to close: −0.744 per unit of the scaled feature, that is, −0.186 logit per additional lane. All four models describe the same association. We do not claim a KAN-specific interpretability advantage. Two earlier analyses are withdrawn: one compared partial-dependence curves with conditional means, which are different estimands, and one ran a bootstrap that split true scenario families.

S-VI. Probes of the Inherent Limitations

Three limitations of the study cannot be removed: it is a simulation, its evaders are scripted, and its vehicles obey SUMO's kinematics. Three registered probes test how much the main contrasts depend on them. None uses a TEST city.

Kinematic model. On fresh scenarios of the seven development-phase cities (avoidant and random evaders, teams of 3, 5 and 7, 20 scenarios and two SUMO seeds), PLB and pursuit were run under three variants with the same scenarios and seeds: the model used throughout (Euler position update, Krauß car-following), a ballistic position update, and IDM car-following. Of 10 080 scheduled episodes, 252 failed scenario placement, in 42 conditions that fail identically for both controllers and all variants (273 families per variant). A development check confirmed beforehand that both options change the simulated trajectories. Both registered primaries were met (Table S-XII). OC rates under the two alternatives differ from the reference by at most 0.018, with intervals that include zero. Eight episodes were flagged as collided, all from one scenario and seed and for both controllers; none is a capture.

Background traffic. Earlier in the project, background traffic had been tested only for the pursuit team. The same scenarios and seeds were therefore run again with about 300 random-trip passenger cars per city (3 360 scheduled episodes, 84 placement failures in the same 42 conditions). The reference without traffic is the first row of Table S-XII. The registered primary was

met: with traffic, PLB minus pursuit is +0.094 [0.060, 0.130] (city → family [0.050, 0.137]). Traffic lowers capture by PLB by −0.062 [−0.095, −0.030] and by pursuit by −0.029 [−0.063, 0.004]. The advantage observed with traffic is 0.032 smaller than without; a post hoc interval for that difference reaches zero ([−0.066, 0.000]; city → family [−0.077, 0.012]), so a smaller advantage with traffic is not established. PLB estimates arrival times in free flow, which could lower its capture with traffic; this was not tested. The gap between the 5 m rule and OC is unchanged. The traffic is one level of random trips; it is not calibrated demand, and background drivers do not react to the pursuit.

Table S-XII: PLB and pursuit under alternative kinematic models and with background traffic. Same scenarios and seeds in every row. OC capture rates; family-clustered 95% intervals.

Condition	PLB	Pursuit	PLB − pursuit	Body gap 5 m − OC, pursuit
Reference model	0.439	0.313	+0.126 [0.090, 0.164]	+0.619 [0.573, 0.665]
Ballistic position update	0.443	0.303	+0.140 [0.103, 0.177]	+0.631 [0.586, 0.676]
IDM car-following	0.449	0.331	+0.118 [0.079, 0.158]	+0.581 [0.531, 0.630]
Reference model, background traffic	0.377	0.283	+0.094 [0.060, 0.130]	+0.638 [0.593, 0.682]

An evader tuned against PLB. All other evaders are fixed scripts. This study gives the evader the kind of tuning that PLB received. A new evader level scores every continuation by a weighted sum of seven features and takes the best; a cul-de-sac ranks last while an alternative exists. The features use only the pursuers' current positions:

- the road distance from the end of the continuation to the nearest pursuer and to the second nearest, each capped at 1 000 m;
- the number of onward continuations, capped at four;
- the share of pursuers within 300 m of its end;
- whether a pursuer is on the continuation or on one of its continuations;
- whether it is the continuation that a distance-greedy predictor expects;
- a reproducible random number.

With weight one on the first feature and zero elsewhere, the rule makes the same choice as the optimal evader, which a unit test checks in 800 random

states; this reference runs on the same scenarios as the tuned evader. The four existing evader levels were re-run on 140 recorded episodes before and after the source change, with identical outcomes.

The search used the TRAINING cities and one objective, the lowest OC capture by PLB. Thirty-two weight vectors were drawn uniformly in $[-1, 1]$ with a fixed seed. A first rung (four cities, six scenarios, team of five) kept the eight best. A second rung (four cities, 20 scenarios, teams of 3, 5 and 7) selected one. Against the selected evader, PLB captures 0.271 on the second rung, against 0.367 for the reference. The selected weights are -0.87 on the nearest distance, -0.89 on occupied continuations and $+0.87$ on the random term, with the others between -0.14 and $+0.27$: the evader avoids roads that carry a pursuer, is hard to predict, and does not seek distance.

The test used fresh scenarios on the seven development-phase cities (20 scenarios, two SUMO seeds, teams of 3, 5 and 7). Of 4 200 scheduled episodes, 190 failed scenario placement, in 38 conditions that fail identically for all five arms (134 families). Table S-XIII gives the rates. The registered primary was met: PLB minus pursuit against the tuned evader is +0.128 [0.081, 0.178] (city $\rightarrow$ family [0.061, 0.198]), and +0.097, +0.127 and +0.162 for teams of 3, 5 and 7. Relative to the reference, the tuned evader changes capture by PLB by -0.034 [-0.099, 0.031] and by pursuit by +0.010 [-0.039, 0.059]. The reduction seen in the search (0.096) is therefore mostly not reproduced on fresh scenarios. The family is scripted and small; a learned evader or a game-theoretic best response could be stronger.

Table S-XIII: OC capture against an evader tuned against PLB and against the reference (optimal) evader, on the same fresh scenarios. Family-clustered 95% intervals.

Evader	PLB	Junction ambush	Pursuit	PLB − pursuit
Tuned against PLB	0.29	0.25	0.16	+0.128 [0.081, 0.178]
Reference (optimal)	0.32	–	0.15	+0.172 [0.115, 0.232]

S-VII. Reproducibility and Command Bookkeeping

Source recoverability. Packages store source hashes rather than complete source snapshots. Seven dated archives of the sources were taken during the audits. Of the 69 packages that recorded source hashes at launch, 14 are fully recoverable from those archives, including the studies added after the

archives began, and 55 only in part: for the earlier packages, later edits are documented by amendments, but some launch versions cannot be recovered. The map is produced by a script and lists every recorded file. Replays with the historical behaviour reproduced every sealed outcome we re-ran: 1 433 captures, 600 sampled episodes, and 500 episodes sampled from five further packages.

Command bookkeeping. Historical runs recorded orders that had no route, or that failed and fell back to a default stop, as applied. The bookkeeping is corrected. Re-running the full PLB test cohort and the goal-directed cohort with it (8 966 episodes) changed no outcome and no contrast. In 58 of 58 re-runs compared in full, the positions of all vehicles agreed to 1 mm at every 0.1 s step (speed, heading and lane were not compared). A fallback remains an attempted manoeuvre whose exceptions are suppressed; logging it does not show that the manoeuvre succeeded physically. Other historical cohorts were not re-run.

Invalid values. The judge compared speeds without checking that they were finite, so a trace with a non-finite speed could have been judged a capture. The judge now returns "unknown" for a trace that contains a non-finite value, a negative speed or a non-positive body dimension, and such an episode is not counted as a capture. For a trace without such values the verdict is unchanged. No such value was found in the 4.28 million frames of the 1 433 replayed captures or in the 512 reconstructed training episodes. Other historical traces were not stored and could not be scanned.

Log files altered after sealing, and restored. A simulator process that outlives its run keeps the run's log file open. When such leftover processes received a termination signal, each wrote one line to the log of a package that had already been sealed, in 14 packages. In every case the sealed content was intact as a prefix of the file and the added bytes were only that line, repeated. The 14 files were cut back to their sealed length, which restores their recorded checksums; the added bytes are kept, with a manifest. No result file was affected. Until then a seal stated that every writer of the package had ended without checking it. The sealing tool now asks the operating system whether any process holds a file of the package open for writing, refuses to seal if one does, and records the check in the seal; the statement in earlier seals is the operator's declaration only. A check of all 89 sealed packages after the leftover processes were ended found no process holding any of their files.

S-VIII. Registration, Addenda and Counting

Counting. The count is produced by a script with one explicit rule per package and block (`scripts/count_episodes.py`); it replaces figures that earlier versions had assembled by hand. In our setting the studies behind the article completed 197 655 evaluation episodes, each run to estimate a rate or a contrast that the article reports. A further 16 403 episodes were used to select parameters or candidates (the search rungs and search validation of PLB, of the seven-pursuer sensitivity study and of the tuned evader) and are not evaluation data. A further 22 455 runs were replays of earlier episodes for audits, and 6 483 scheduled tasks failed, almost all because the scenario generator could not place the team; such failures affect all methods of a condition identically. Episodes generated while training a learner, validation during training and development checks are not counted; the 499 training episodes replayed to audit the training loop are counted as replays. The audits of the two published environments add 2 484 episodes and 1 709 replays in those environments. These are counts of simulations. Episodes of one scenario family are not independent observations, which is why every interval in the article resamples families.

Registered and post hoc analyses. Table S-XIV lists the studies behind the article and whether their primary analysis was registered before the runs.

Table S-XIV: Studies behind the article. "Registered" means that the protocol and the primary analysis were recorded, and sources hashed, before the runs. Outcomes are stated for the primary analysis.

Study	Registered	Primary analysis	Outcome
Endpoint robustness (three evaders, team sizes, grids)	yes	team minus independence reference	no supported excess in 10 cells
Junction ambush	yes	ambush minus pursuit	+0.04 to +0.12
Option PPO (MLP, KAN)	yes	learned minus rule	0.000 [−0.004, 0.004]
PLB parameter search and test	yes	PLB minus junction ambush	+0.050, met

Study	Registered	Primary analysis	Outcome
Oracle evader model	no	descriptive	+0.03 to +0.04, intervals include zero
Pursuer speed	no	descriptive	−0.07 to +0.03
First MAPPO study (first implementation)	yes	MAPPO minus planner	met on its own test (+0.047 MLP, +0.052 KAN); see Table S-X
Decision-commitment replication	yes	3 s minus 1 s commitment	+0.010 [−0.012, 0.033], not confirmed
Hold audit of OC	yes	escapes with pursuers held	0 of 1 433 after follow-up completion
Held-out evaluation (TEST)	yes	PLB minus pursuit; proximity minus OC	both met
Goal-directed evader	yes	PLB minus pursuit	+0.105, met
MAPPO-v2 (first implementation)	yes	v2 minus PLB	not met; fine-tuning affected by an implementation defect
Coordination ablation	yes	coordinated minus independent PLB	+0.071, met
Active-escape probe	yes	descriptive	15 of 1 433 escapes
Null team; joint escape; clustered intervals	yes (as corrections)	descriptive	see article
Proximity-trained MAPPO (first implementation)	yes	widening of the gap between endpoints	+0.170, met; not reproduced after correction (see MAPPO-v3)
MAPPO-v3 (corrected controller)	yes	validated gain on imitation; widening of the gap under proximity training	neither met; the earlier decline did not recur; training loop later found defective
Second published environment (PEPCRL-MVP)	yes	descriptive, with a registered prediction	no same-lane capture in 2 380 events; prediction held
Second held-out set (HELD-OUT B)	yes	PLB minus pursuit; proximity minus OC	both met (+0.064; +0.549)

Study	Registered	Primary analysis	Outcome
Comparison of the two held-out sets	no	post hoc	advantage of PLB smaller on the second set
Capture instants of the hold audit, re-verified	yes (as correction)	descriptive	1 433 of 1 433 confirmed; none affected
First training batches of MAPPO-v3, reconstructed	yes (as correction)	descriptive	8 of 8 batches reproduced; 0.6% of trained decisions had no future
MAPPO-v4 (corrected training loop)	yes	validated gain on imitation; widening of the gap under proximity training	neither met; conclusions of MAPPO-v3 unchanged
Published environment, addendum	yes (as correction)	descriptive	12 of 12 follow-ups and 800 of 800 re-runs reproduce the original events
Published environment (DGMARL-MVP)	yes	descriptive	no same-lane capture in 999 events
Kinematic model (ballistic update; IDM)	yes	PLB minus pursuit under each	+0.140 and +0.118, both met
Background traffic	yes	PLB minus pursuit with traffic	+0.094, met
Evader tuned against PLB	yes	PLB minus pursuit against it	+0.128, met
KAN, spline, MLP, logistic models	no	post hoc	no KAN advantage detected
Lane-duplication diagnostics	no	post hoc	49% versus 0% duplicated targets

The first MAPPO study met its primary on its own test set, with a gain over the planner of about 0.05. Later results are mixed (Table S-X): on the held-out cities the same controllers show no detected difference from PLB, the MAPPO-v2 study did not meet its primary, a post hoc contrast on a further test set is again positive, and the MAPPO-v3 and MAPPO-v4 test

sets give differences whose intervals include zero. A registered replication of one suspected cause, a 3 s decision commitment, was not confirmed. In every study the retained controllers are imitation checkpoints, so the article attributes no gain to reinforcement learning and leaves the cause of the small difference open. Earlier phases of the project (constructive caging certificates and reinforcement-learning pilots with a gate heuristic), closed before the studies above, are not used in the article; their records are part of the repository.

Versioned corrections. Six addenda corrected earlier analyses without relabelling original outputs:

- *Addendum 1* completed the 60 s follow-up of 188 censored hold-audit replays, introduced joint escape (gate or network exit) for the containment indicators, and replaced cell-level bootstraps by family-clustered and city → family bootstraps. The article reports the pre-registered cell interval of the held-out primary together with the corrected ones.

- *Addendum 2* verified each extended replay against its original capture, added family intervals for the null team and the indicators, recomputed the KAN comparison with true scenario families, and corrected the command bookkeeping.

- *Addendum 3* added the coordination ablation, the active-escape probe and the large-scale re-runs with the corrected bookkeeping. One runner of this addendum was changed during the run to add a stall timeout; the change is documented in the package and does not alter any episode.

- *Addendum 4* followed a fourth audit. It completed the follow-up of 12 censored cases in the published-environment audit and replaced its stop analogue by one with a duration measured in simulation time (Section S-II); it identified the implementation defect of the learning studies and repeated them with a corrected controller (Section S-IV); and it added a post hoc interval for the effect of traffic on the advantage of PLB.

- *Addendum 5* followed a fifth audit. It replayed the 1 433 audited captures to verify each capture instant with a corrected extractor (Section S-I); it made the judge reject invalid values (Section S-VII); it corrected the training loop and measured the incidence of its two defects on the first training batch of every MAPPO-v3 run, and then

repeated the learning block with the corrected loop (Section S-IV); it withdrew the attribution of the earlier decline under fine-tuning to one defect; it recorded an arithmetic erratum of a registered protocol; and it added an antecedent that the article had omitted [4].

- *Addendum 6* followed a sixth audit and ran no simulation. It stated the denominators of the second published environment, with the 22 capture events recorded before an abort (Section S-II); it made the sealing tool check that no process still writes to a package (Section S-VII); it replaced "no deterioration" by a wording that does not suggest a demonstrated equivalence and separated training attempts from valid episodes (Section S-IV); and, after the full text of [4] had been read, it corrected the description of that study in the article.

References

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